Exp: 07

Date:

Develop a simple calculator using XML-RPC

Aim:

To develop a simple calculator using the XML-RPC (XML Remote procedure call) protocol in Python to perform basic arithmetic operations between a client and a server.

Requirements:-

Hardware:

- Laptop
- minimum 4 gb Ram

Software:

- Windows 10/11
- Python 3.8 or above
- VS Code

Procedure:-

1. Download Python.
2. Install Python
3. Select Add Python to PATH during Installation.
4. Click Install now
5. Wait until the Installation complete.

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Code:

```
from xmlrpc.server import SimpleXMLRPCServer

server = SimpleXMLRPCServer(("localhost", 8000))

Print ("XML-RPC calculator server Running on Port 8000....")

# Functions

def add(a,b):
    return a+b

def subtract(a,b):
    return a-b

def multiple(a,b):
    return a*b

def divide(a,b):
    if b == 0:
        return "Division by zero is not allowed"
    return a/b

# Register Function

server.register_functions(add, "add")

server.register_functions(subtract, "subtract")
```

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```
server.register_function(multiple, "multiply")
server.register_function(divide, "divide")

# Start server

server.serve_forever()
```

save the file.

Write the client program

Open client.py and type the following code.

```
import xmlrpc.client

server = xmlrpc.client.ServerProxy("http://localhost
                                   :8000")
a = int(input("Enter First number : "))
b = int(input("Enter second number: "))
Print("\n select operation")
Print("1. Addition")
Print("2. subtraction")
Print("3. multiplication")
Print("4. Division")

choice = input("Enter your choice (1-4) : ")
if choice == '1':
    Print("Result =", server.add(a,b))
```

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```
elif choice == '2':
    print ("Result =", server.subtract(a,b))

elif choice == '3':
    print ("Result =", server.multiply(a,b))

elif choice == '4':
    print ("result = ", server.divide(a,b))

else
    print ("Invalid choice")
```

save the file.

Result:- Thus, a simple calculator was successfully developed using XML-RPC. The client sent arithmetic operation requests to the server, and the server performed the calculations and return the result to the client.

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